CS230: Requirements Engineering
School of Information Technology, Engineering, Mathematics and Physics

Final Examination
Semester 1 2023

F2F Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 3 hours

Total marks: 100

Instructions:

1. *This exam has three sections, with a total of 11 questions.*
2. *Section A 30 marks, Section B 40 marks, Section C 30 marks.*
3. *This exam is worth 50 % of your overall mark.*
4. *The minimum pass mark for this exam is 40 out of 100 marks*
5. *There are 18 pages in this paper including this cover page.*
6. *This is a closed book Exam.*
7. *This is a question and answer booklet.*
8. *For the extra sheet, start each question on a new page.*
9. *All the questions are compulsory. There are no choices.*

Student Name: _____

Student ID: _____

~ALL THE BEST~

Section A: Short Answers (30 Marks)

Question 1 **(5 marks)**

a) Explain the importance of requirements engineering in software development and the consequences of not performing it effectively (2 marks)
The process of establishing the services that the customer requires from a system and the constraints under which it operates and is developed. ensures a clear understanding of the customer's needs and desires, helps mitigate risks of vagueness and ambiguity, and leads to the creation of secure and successful software products

b) Differentiate between information acquisition and requirements elicitation? (2 marks)
Information acquisition is finding information about the current system whereas requirements elicitation is identifying the needs of the stakeholders

c) Who is a stakeholder for a system? (1 mark)
Stakeholders are the people who have an interest or influence on the system, such as users, customers, sponsors, managers, regulators, or suppliers. They provide the requirements, expectations, feedback, and resources for the system. End users.

Question 2 **(5 marks)**

a) What is a domain requirement and explain one domain requirement problem? (2 marks)
Domain requirements may be new functional or non-functional requirements, constraints on existing requirements, or define specific computations. If domain requirements are not satisfied, the system may be unworkable. Two main problems with domain requirements: Understandability.

Understandability - Requirements are expressed in the language of the application domain. This is often not understood by software engineers developing the system. Implicitness - Domain specialists understand the area so well that they do not think of making the domain requirements explicit. This is major problem in communications.

b) List six stakeholders for a healthcare patient information system. (3 marks)
Patients, Doctors, Receptionist, Vendors, Government, Policy makers

Question 3 **(5 marks)**

a) What is system re-engineering, and give two main reasons for undertaking it? (3 marks)

Improved efficiency: By analyzing and redesigning processes, re-engineering can lead to significant improvements in productivity, speed, and cost-effectiveness. Increased flexibility: Re-engineering can make systems more adaptable to changing business needs and market conditions.

It is restructuring or rewriting part or all legacy system without changing its functionality.
Reduced Risk – high risk in new software development
Reduced Cost – cost of re-engineering is less compared to the cost of developing a new software

b) What is prototyping, and how can it be used as a verification and validation technique for requirements? (2 marks)
prototyping serves as a valuable technique for verifying and validating requirements by providing a tangible representation of the system, enabling early feedback, and facilitating requirement evolution.

Question 4 **(10 marks)**

a) Difference between the goal of Elaboration and Negotiation? (2 marks)

Elaboration goals – create models such as ER diagrams, Use cases, DFD that defines information and function.

Negotiation goal is to resolve requirements that are conflicting, costly and unrealistic. Negotiate their win conditions into win-win conditions for everyone.

b) Why we should find a possible way to eliminate the need for the meeting? (1 mark)

Lacks agenda and uses the company resources such as time which could be utilized else where.

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c) Write the type of hierarchy is this "Animal classification: the set of 'birds' contains the set of 'birds of prey' which contains the set of 'eagles' which contains the set of 'golden eagles'". (1 mark)

part of

d) Explain the term Requirements Reviews? (1 mark)

an opportunity to check the requirements document for errors and omissions and confirm that the overall quality of the requirements is acceptable before the project proceeds.

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e) What is the difference between requirement specification and validation? (2 marks)

User Requirements Specifications are written early in the validation process, typically before the system is created

Requirements Validation

Check that the right product is being built

Ensures that the software being developed (or changed) will satisfy its stakeholders

Checks the software requirements specification against stakeholders goals and requirements

f) List An example of an ambiguous functional system requirement is:
"The system should be able to handle a large amount of data"
What issue is caused by such a requirement and write a correct example to make this unambiguous? (3 marks)

The example requirement "The system should be able to handle a large amount of data" is **ambiguous** because it lacks specificity. It does not define what "large amount of data" means in practical terms, leading to potential misunderstandings and implementation challenges.

To make this requirement **unambiguous**, we need to provide clear criteria. Here's a corrected example:

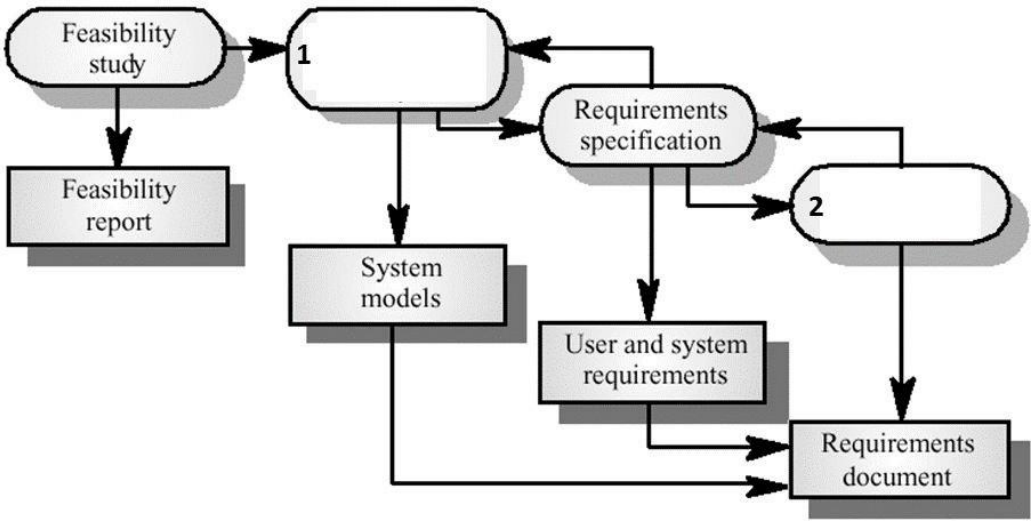
Corrected Requirement: "The system must support concurrent processing of at least 10,000 user requests per minute while maintaining response times below 500 milliseconds."

In this revised requirement:

- We specify a measurable metric (10,000 user requests per minute).
- We set a performance threshold (response times below 500 milliseconds).
- The requirement is now precise and actionable, allowing developers to design and test accordingly.

Question 5 **(5 marks)**

a) Write the missing process from the requirement engineering process at space provided at 1 and 2. (2 marks)



1 – Requirements elicitation and analysis 2 – Requirements validation

b) Why are software changes inevitable? Explain one of the main factors that lead to software changes and provide an example to illustrate your points. (3 marks)
Systems are tightly coupled with their environment. When a system is installed in an environment it changes that environment and therefore changes the system requirements. Systems MUST be changed if they are to remain useful in an environment.

Software changes are inevitable due to the dynamic nature of technology, evolving business requirements, and the need to adapt to new environments. Here are some reasons why software changes occur:

1. **Emergence of New Requirements**:
    - As software is used, new requirements emerge.
    - Business needs evolve, and users discover additional features they want.
    - Example: A banking application initially designed for basic transactions may need to incorporate mobile banking features due to changing customer preferences.

Example:

Consider an airline reservation system. Initially, it was designed for booking flights, managing passenger data, and issuing tickets. However, over time, new requirements emerged:

1. **Mobile Check-In**:
    - As mobile usage increased, passengers demanded the ability to check in via mobile apps.
    - The system needed updates to accommodate mobile check-in features.
2. **Loyalty Programs**:
    - Airlines introduced frequent flyer programs.
    - The reservation system had to integrate loyalty points tracking and redemption features.
3. **Security Enhancements**:
    - With rising cybersecurity threats, the system required stronger authentication and encryption.
    - Regular security updates became essential.
4. **API Integration**:
    - Airlines partnered with hotels, car rental services, and travel agencies.
    - The system needed APIs to connect with external services for seamless travel experiences.

In each case, software changes were necessary to meet evolving requirements and maintain a competitive edge in the airline industry.

End of Section A

Section B: Long Answers (40 Marks)

Question 6 **(6 marks)**

Purchasing System

The Purchasing Department handles purchase requests from other departments in the company. People in the company who initiate the original purchase request are the "customers" of the Purchasing Department. A case worker within the Purchasing Department receives the request and monitors it until it is ordered and received. Case workers process requests for the purchase of products under $1,500, write a purchase order, and then send it to the approved vendor. Purchase requests over $1,500 must first be sent out for bid from the vendor that supplies the product. When the bids return, the case worker selects one bid and then writes a purchase order and sends it to the vendor.

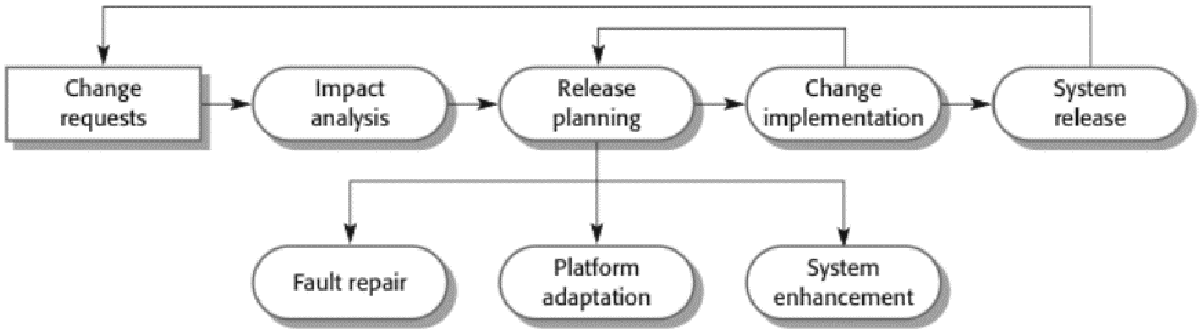
Develop an Activity diagram based on the above case study. Note any ambiguities or questions that you have as you develop the model. If you need to make assumptions, also note them. (6 marks)

Question 7 Software Evolution and Requirement Management (6 Marks)

a) Write two tasks that need to be taken care of while correcting a fault in the Implementation phase rather than correcting the same fault in the Planning phase? (2 marks)

1. **Code Modification and Debugging**:
   - **Implementation Phase Task**:
     - Identify the specific code segment responsible for the fault.
     - Modify the code to fix the issue.
     - Debug the modified code to ensure correctness.
   - **Reason for Implementation Phase**:
     - During implementation, direct access to the codebase allows efficient code changes and immediate testing.
     - Debugging tools and real-time execution aid in identifying and rectifying faults.
2. **Integration Testing and Verification**:
   - **Implementation Phase Task**:
     - Integrate the modified code with the existing system.
     - Conduct integration testing to verify that the fault is resolved.
     - Validate that the fix does not introduce new issues.
   - **Reason for Implementation Phase**:
     - Integration testing requires the actual system components to be in place.
     - Verifying the fix within the complete system context ensures overall functionality.

b) Refer to the diagram given below. How is change implementation viewed in this case? (2 marks)



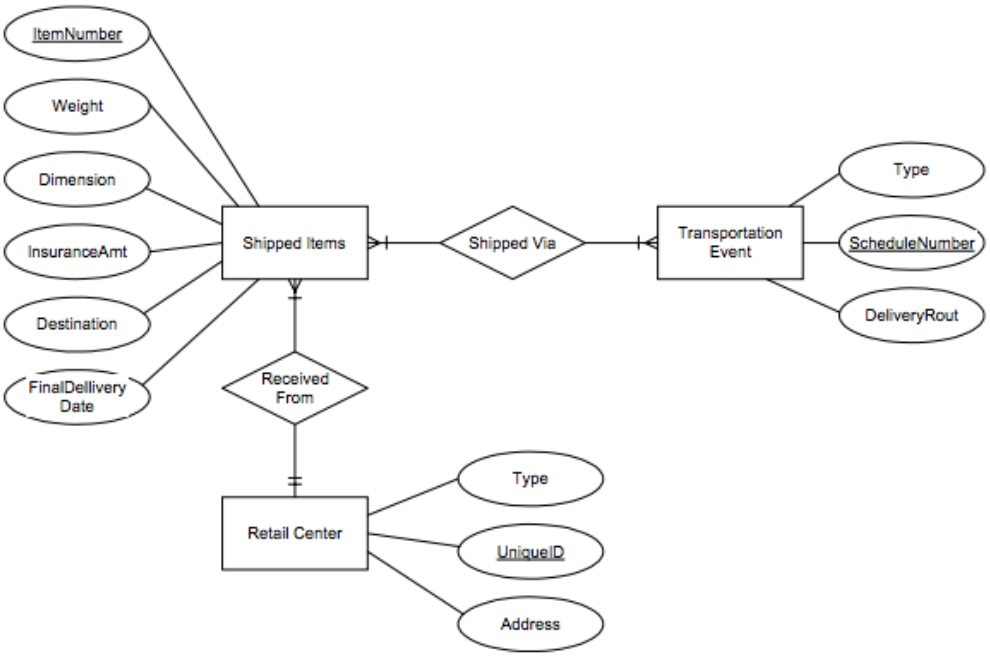
c) What is version control in Requirements Management Activities and why is it important? (2 marks)

Version control, also known as source control, is the practice of tracking and managing changes to software code. Version control systems are software tools that help software teams manage changes to source code over time.

Question 8 ERD **(7 marks)**

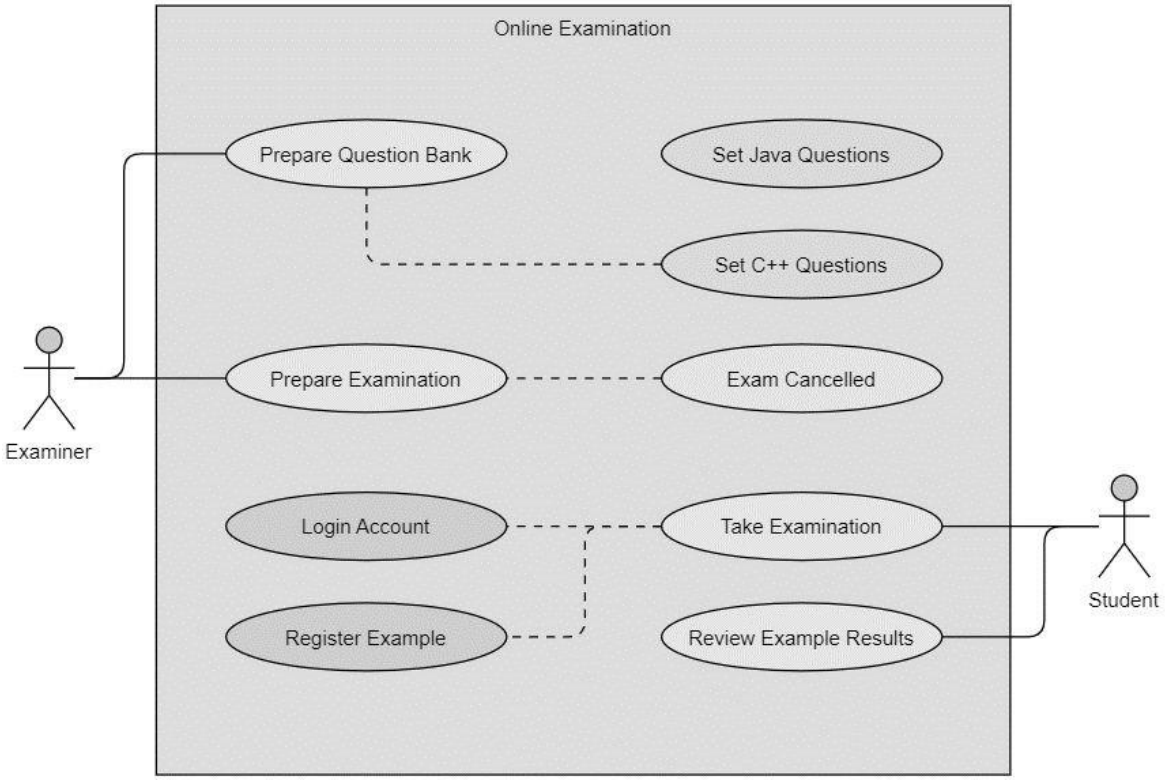
UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute.

Please create an Entity Relationship diagram that captures this information about the UPS system. Be certain to indicate identifiers and cardinality constraints. (7 marks)

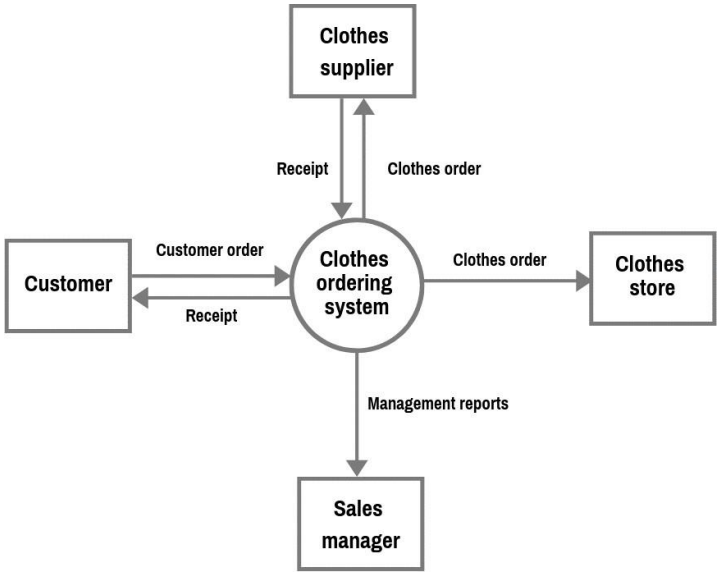


Question 9 UML **(9 marks)**

a) Use the online examination use case below and draw the use case relationship between the use cases. Clearly with correct flow show the type of use case relationship – Generalization, Include, Extend. (2 marks)



b) The Context Diagram represents the entire information system as one big process.



A simple context DFD drawn for a Clothes Ordering System is given above. Draw a DFD level 1 diagram for the system. **(7 marks)**

Question 10 Preparing a feasibility report and software requirements (12 marks)

You are required to read the following system specifications and answer the questions below.

Tourist management system

Bula Tourism would like to invest in new system that is a tourist management system that will be used to manage tourist activities, such as hotel bookings, restaurant reservations, and tour bookings. The system will also be used to track tourist information, such as demographics, travel patterns, and preferences.

The system will be designed to be user-friendly and accessible to tourists and local businesses alike. The system will be available online and will be accessible from any device with an internet connection. Tourists will be able to use the system to plan their trips, make reservations, and book tours. Local businesses will be able to use the system to manage their bookings and track tourist information.

The system will include the following features:

- Hotel Booking: Tourists will be able to search for and book hotels through the system. They will be able to view hotel ratings and reviews, as well as room availability and pricing.
- Restaurant Reservations: Tourists will be able to search for and reserve restaurants through the system. They will be able to view restaurant ratings and reviews, as well as menus and pricing.
- Tour Booking: Tourists will be able to search for and book tours through the system. They will be able to view tour descriptions, pricing, and availability.
- Tourist Information: The system will track tourist information, such as demographics, travel patterns, and preferences. This information will be used to personalize the tourist experience and improve the tourism industry in the country.
- Feedback and Reviews: Tourists will be able to provide feedback and reviews of hotels, restaurants, and tours through the system. This feedback will be used to improve the quality of services provided by local businesses.

The system will be developed using Agile software development methodologies and will be tested extensively to ensure that it is user-friendly and meets the needs of both tourists and local businesses. The system will also be designed to be scalable, to accommodate the expected growth in tourism in the country over the next few years.

Prepare a feasibility report for the system specifications. The report should contain the following sections:

a) Description of the current situation (1 mark)

The current situation in the tourism industry lacks an efficient and centralized system for managing tourist activities, tracking tourist information, and facilitating interactions between tourists and local businesses. Existing processes are manual, fragmented, and do not fully meet the needs of both tourists and service providers.

b) Explain the benefits of the system (2 marks)

1. Enhanced Tourist Experience:
   - The system streamlines hotel bookings, restaurant reservations, and tour bookings.
   - Tourists can plan their trips seamlessly, access real-time information, and personalize their experiences.
   - Improved convenience and satisfaction lead to repeat visits and positive word-of-mouth.

2. Efficient Business Operations:
   - Local businesses can manage bookings, track tourist preferences, and optimize resource allocation.
   - Automation reduces administrative overhead, minimizes errors, and enhances service quality.
   - Businesses gain insights into customer behavior for targeted marketing.

c) Two anticipated risks during system development (2 marks)

1. **Integration Challenges**:
   - Integrating with existing hotel, restaurant, and tour booking systems may be complex.
   - Data migration, API compatibility, and synchronization pose risks.
   - Mitigation: Rigorous testing and phased integration to minimize disruptions.
2. **Scalability Issues**:
   - As tourism grows, the system must handle increased user load.
   - Scalability challenges may arise in terms of performance, database capacity, and concurrent access.
   - Mitigation: Design for scalability, monitor system performance, and plan for infrastructure upgrades.

Identify the requirements for the system as listed above tourist management system:

d) Any three functional requirements (3 marks)

1. **Hotel Booking**:
   - Tourists can search for hotels.
   - View ratings, reviews, availability, and pricing.
   - Book rooms online.
2. **Restaurant Reservations**:
   - Tourists can search for restaurants.
   - Reserve tables.
   - Access menus, ratings, and reviews.
3. **Tour Booking**:
   - Tourists can search for available tours.
   - View descriptions, pricing, and availability.
   - Book tours.

NOTE: The format to write this

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e) Any two Non-functional requirements (2 marks)

1. **Usability**:
   - The system must be intuitive and user-friendly.

   - Clear navigation, responsive design, and minimal learning curve.
2. **Scalability**:

   - The system should handle increasing user traffic.
   - Accommodate growth in both tourists and local businesses.

f) Identify any four stakeholders (2 marks)

1. **Tourists**: End-users who interact with the system for trip planning and bookings.
2. **Local Businesses (Hotels, Restaurants, Tour Operators)**: Service providers using the system for managing bookings and tracking tourist data.
3. **IT Admins and Developers**: Responsible for system maintenance, security, and enhancements.
4. **Tourism Authorities and Government Agencies**: Interested in data analytics and industry insights.

End of Section B

Section C: Case Study (30 Marks)

Question 11 Scenario-based questions.

Read the problem statement given below very carefully and use it to answer the questions that follow.

You are required to read the following system specifications and answer the questions below.

PacifiCars Mobile app

PacifiCars is a car rental company and is planning to develop a mobile app that will allow registered customers to reserve cars online. The payment is handled when the customer comes to pick up the car but is outside the scope of the system.

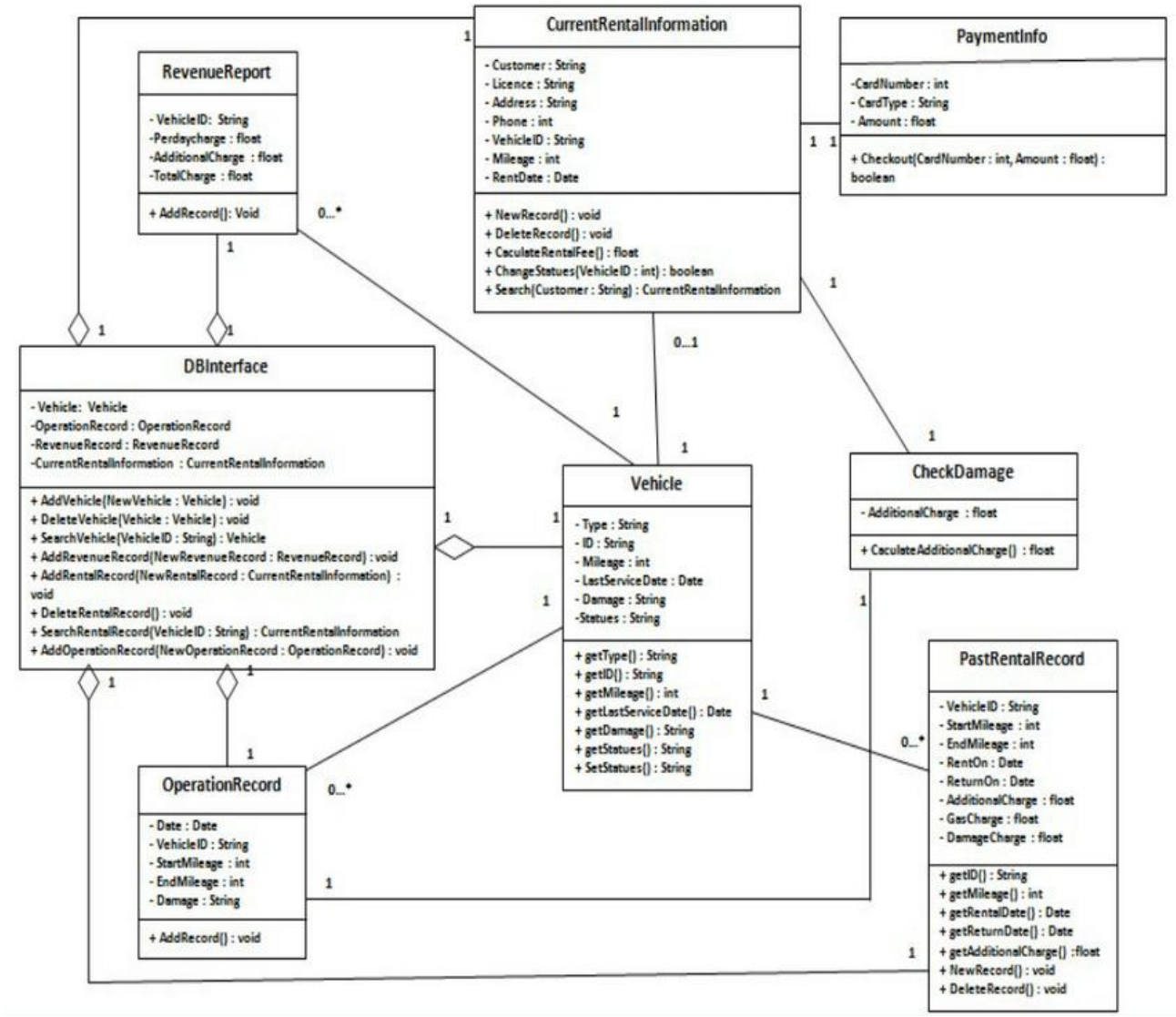
Before a customer can submit a reservation request for a car, the customer has to register online using the system. The customer fills out an online registration form and submits using the system. The Sales Manager approves the registration request online, and upon approval, the system generates and emails the customer the login details. The customer can also update their profile, for example, change their email address, name, etc. For each customer, the company needs the name, residential and postal address, email, and home and work phone numbers.

A registered customer can search for cars. When searching for cars the customer can provide the date range (that is when the customer wants to pick the car and when the car will be returned to the company), the car type, etc. After choosing a particular car, the customer can submit a reservation request for that car. This reservation request is reviewed and approved by the Sales Manager using the system. The Sales Manager's decision (whether the application was approved or denied) is emailed to the customer. The basic data collected about online reservations is the Date, customer's details, car details, pick-up date, return date, and approval date.

The Inventory Manager is responsible for adding, removing, or updating the car details. The company records the car registration, description of the car, and the last recorded mileage. The mileage is updated every time the car is returned by the customer. Apart from online booking, the company's salespersons can also reserve cars directly using the system. In this case, the reservation does not require approval from the Sales Manager. The reservation details recorded by the system include the Date, customer's details, car details, pick-up date, return date, and salesperson details.

For the same PacifiCars mobile app case, develop the diagrams in (a) to (d):

a) Convert your ERD to a domain class diagram (7 marks)



b) Create a detailed use case diagram for the PacifiCars mobile app capturing the functional requirements. (10 marks)

c) Using the above description, write a fully developed description for use cases: **Manage Cars,** use case scenario showing the Brief Description, Actors and the Flow of Events (FoE)/Activities, preconditions, postconditions exception conditions etc. For the FoE show the Actor and System tasks. (7 marks)

d) Now create a System Sequence diagram for **Search Cars** (6 marks)

THE END